

2025

COMPUTER SCIENCE

Paper : CSME-402

(Elective-II : Distributed and Cloud Computing)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer *question nos. 1 & 2* and *any four* from the rest.1. Answer *any five* questions :

2×5

- (a) Mention the characteristics of the multi-cloud system.
- (b) What is a container? Give an example.
- (c) What is a hypervisor? Give an example.
- (d) What do you mean by open cloud ecosystem?
- (e) What is IAM in cloud? What are the challenges in IAM?
- (f) What do you mean by Multi-tenant design?
- (g) Explain the necessity of VM migration.

2. Answer *any five* questions :

4×5

- (a) Describe the concept of Kubernetes and its key components.
- (b) Compare and contrast the storage, memory and data virtualization.
- (c) How do different service models (IaaS, PaaS, SaaS) cater to varying business needs?
- (d) Discuss the concept of scalability in cloud computing. How is it different from elasticity?
- (e) Explain the techniques of object storage as implemented in Amazon S3.
- (f) Mention different cloud services offered by Amazon Web Services (AWS). Illustrate Amazon EC2 services and its basic features.
- (g) Explain the term 'security' in a hybrid cloud. Define virtual machine security.

3. (a) Compare and contrast the four primary cloud deployment models: public, private, community, hybrid and multi-clouds.
- (b) How do we perform live migration of Virtual Machine?

6+4

Please Turn Over

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4. (a) Explain the characteristics of Amazon S3. 4+6
(b) How does Cross Region Replication work in Amazon S3? 6+4
5. (a) What are the different ways to purchase EC2 instances? Discuss their relative advantages and disadvantages. 5+5
(b) Explain step-by-step process to set up an EC2 instance in AWS. 6+4
6. (a) How does one address the issues of under-provisioning and over-provisioning in traditional computing models? 5+5
(b) Explain the concept of 'fault tolerance' and its importance in cloud services. 5+5
7. (a) Differentiate between full virtualization and para virtualization. 5+5
(b) Explain the specialized features that influence IaaS offerings. 5+5
8. (a) How do security and privacy affect the entire Cloud Computing stack? 5+5
(b) What is SLA? Explain about life cycle of SLA. 5+5
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